

① prove $A + A'B = A + B$

$$A + A'B \Rightarrow (A + A') \cdot (A + B) = 1 \cdot (A + B) = A + B$$
$$(A + A' = 1)$$

② prove $(A+B)(A+C) = A + BC$

$$(A+B)(A+C) = \underbrace{A \cdot A}_A + A \cdot B + A \cdot C + B \cdot C$$
$$= A(1 + \underbrace{B+C}) + BC \quad (1 + X = 1)$$
$$= A(1) + BC$$
$$= A + BC$$

③ prove $(A+B)' = A'B'$

For this we need to show that $A+B$ and $A'B'$ satisfy complement laws $(X + X' = 1 \quad X \cdot X' = 0)$

$$(A+B) + (A'B') = 1$$

$$\Rightarrow ((A+B) + A') \cdot ((A+B) \cdot B')$$
$$= (A + A' + B) \cdot (A + B + B')$$
$$= (1 + B) \cdot (1 + A)$$
$$= 1 + \underbrace{B + A + A \cdot B}_X = 1 + X = 1 \quad \checkmark$$

$$(A+B) \cdot (A'B') = 0$$

$$\underbrace{A A' B'}_0 + \underbrace{B B' A'}_0 = 0$$

$$\underbrace{0 + 0}_{0} = 0 \quad \checkmark$$

Since both complement laws are satisfied

$$(A+B)' = A'B'$$

$$(4) AB + AB' = A$$

$$A(B+B') = A(1) = A \checkmark$$

$$(5) \text{ prove } A \oplus B = AB' + A'B$$

this is XOR gate $\rightarrow$ output high only when inputs are different (one is true other false)

$$\therefore A \oplus B = AB' + A'B$$

same for XNOR gate.

$$(6) A + AB = A$$

$$A(1+B) \\ = A(1) = A \checkmark$$

$$(7) \text{ prove } A(A+B) = A$$

$$\underline{A \cdot A} + AB$$

$$A + AB$$

$$= A(1+B) = A(1) = A \checkmark$$

$$(8) \text{ prove } (A+B')(A'+B) = AB + A'B'$$

$$= \underbrace{AA'}_0 + A'B' + AB + \underbrace{BB'}_0$$

$$= AB + A'B' \checkmark$$

① prove $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$

$$\rightarrow (A+B)(A'+C)(B+C+AA')$$

$$\rightarrow \underbrace{(A+B)}_{\sim} \underbrace{(A'+C)}_{\sim} \underbrace{(B+C+AA')}_{\sim}$$

$$A+B \underbrace{(1+C)}_1 A'+C \underbrace{(1+B)}_1$$

$$\Rightarrow (A+B)(A'+C) \checkmark$$

② $AB + A'C + BC = AB + A'C$

$$AB + A'C + BC(A+A')$$

$$= AB + A'C + ABC + A'BC$$

$$= AB \underbrace{(1+C)}_1 + A'C \underbrace{(1+B)}_1$$

$$= AB + A'C \checkmark$$